

Standalone value of context-specific knowledge

1. Baseline objects

$$P_G = G(X), \quad P_I = G(Y)$$

$$Y = \nabla^{-2} + \lambda_I e + \tau_A$$

$$\Delta G = f(1,0) - f(0,0)$$

$$\Delta I = f(0,1) - f(0,0)$$

$$\Delta X = f(1,1) - f(1,0) - f(0,1) + f(0,0)$$

2. Four productions outcomes

	$A_I = 0$	$A_I = 1$
$A_G = 0$	f_{00}	$f_{00} + \Delta_I$
$A_G = 1$	$f_{00} + \Delta_G$	$f_{00} + \Delta_I + \Delta_G + \Delta_X$

$$\begin{aligned} E[F] = & f_{00} (1 - P_G) (1 - P_I) \\ & + (f_{00} + \Delta_G) P_G (1 - P_I) \\ & + (f_{00} + \Delta_I) (1 - P_G) P_I \\ & + (f_{00} + \Delta_G + \Delta_I + \Delta_X) P_G P_I \end{aligned}$$

$$E[F] = f_{00} + \Delta_G P_G + \Delta_I P_I + \Delta_X P_G P_I$$

Therefore,

$$U(e; X, \tau_A) = f_{00} + \Delta_G G(X) + [\Delta_I + \Delta_X G(X)] G(Y) - \frac{e^\lambda}{\lambda}$$

3. Human effort FOC

$$U_e = [\Delta_I + \Delta_X G(X)] \lambda_I g(Y) - e^{\lambda-1}$$

Hence, for an interior optimum,

$$[\Delta_I + \Delta_X G(X)] \lambda_I g(Y) = e^{\lambda-1}$$

with

$$g(Y) = G'(Y) > 0, \quad g'(Y) < 0$$

And:

$$U_{ee} = [\Delta_I + \Delta_X G(X)] \lambda_I^2 g'(Y) - (\lambda-1) e^{\lambda-2} < 0 \Rightarrow \text{unique optimum}$$

4. Paper baseline : $\Delta_I = 0$

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$$\chi = 0$$

we have

$$G(0) = 0$$

Therefore,

$$U_e = -e^{\alpha-1}$$

Hence,

$$e^*(0, \tau_A) = 0$$

So zero general knowledge implies zero human learning effort

5. Our proposed extension: $\Delta_I > 0$

Maintain

$$\Delta_I > 0, \quad \Delta_X > 0, \quad \Delta_G = 1 - \Delta_X - \Delta_I \geq 0$$

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$$\chi = 0,$$

the FOC becomes

$$\Delta_I \lambda_I g(\bar{v}^{-2} + \lambda_I e + \tau_A) = e^{\alpha-1}$$

Under

$$\Delta_I > 0, \quad \lambda_I > 0, \quad \alpha > 1,$$

and finite $\bar{v}^{-2}$ and τ_A ,

$$e^*(0, \tau_A) > 0$$

Interpretation:

$\Delta_I > 0 \rightarrow$ context-specific knowledge has standalone productive value
 $\rightarrow$ humans still have an incentive to learn even when $\chi = 0$

6. Implication for public knowledge

The baseline transition law is

$$\chi_{t+1} = [(\chi_t + \lambda_G I e^*)' + \varepsilon^2]'$$

At $\chi=0$

$$F_{\Delta_I}(0) = [(\lambda_G I e^*(0, \tau_A))' + \varepsilon^2]^{-1} > 0$$

provided

$$\lambda_G > 0, \quad I > 0$$

7. Final result

$$\Delta_I > 0 \rightarrow F_{\Delta_I}(0) > 0$$

Result. Positive standalone value of context-specific knowledge removes the exact zero-knowledge fixed point for finite AI precision

It doesn't imply that knowledge collapse disappears in general